

ResearchMind Architecture and Retrieval Specifications

ResearchMind utilizes Reciprocal Rank Fusion (RRF) to combine dense vector search with sparse BM25 keywords.

Section 2. Layout Bounding Box Normalization

All PyMuPDF coordinates are normalized into percentage ratios relative to page dimensions:

$\text{norm_x0} = x0 / \text{page_width}$, $\text{norm_y0} = y0 / \text{page_height}$.

Section 3. FlashRank Cross-Encoder Reranking

Top candidate chunks pass through FlashRank cross-encoder reranking before prompt synthesis.